

Mathilde AGUIAR

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EDUCATION

- **Laboratoire Interdisciplinaire des Sciences du Numérique** Orsay, France | Oct. 2023 - Oct. 2026
PhD student; at Université Paris-Saclay
Thesis subject: Natural Language Inference for clinical trials
- **Institut des sciences du Digital Management et Cognition** Nancy, France | Sep. 2022 - Sep. 2023
Master 2 in Natural Language Processing; with honors
Courses: speech processing, computational linguistics, text processing, deep neural networks, lexical analysis, syntactic models, terminology and ontology
- **Polytech Grenoble** Grenoble, France | Sep. 2019 - Sep. 2022
Computer science engineer degree; with honors
Courses: Natural Language Processing, human-machine interaction, Object Oriented Programming, signal processing, data science, operation research, databases, probabilities, computer graphics, Management, Mathematics, English

SKILLS SUMMARY

- **Languages:** Python, Java, SQL, Bash, R, LaTeX, HTML/CSS
- **Frameworks:** PyTorch, SpaCy, Flask, SQLite, HuggingFace
- **Tools:** Docker, Git, Elasticsearch, Kibana, SLURM/OAR
- **Platforms:** Linux, Windows, HPC (Jean Zay, ADAstra, Grid5k)
- **Languages:** French (native), English (Linguaskill: C1 or above), Chinese (equivalent to HSK3), Japanese (equivalent to JLPT N4), German (A2)

EXPERIENCE

- **Maison d'Initiation et de Sensibilisation aux Sciences** Orsay, France | Feb. 2025 - Sep. 2026
Doctoral mission
 - **Conception and animation of a workshop on AI for middle school students:** Conception of a 3-day workshop for middle school and high school students to make them discover AI and research. The workshop is especially designed to encourage girls to enroll in STEM studies. Design of activities adapted to teens and aiming to cover basic AI concepts with applications in healthcare and environment.
- **GETALP, LIG** Grenoble, France | March 2023 - Sep. 2023
Internship
 - **Research Internship - Jargon:** Various contributions to the development of a new version of the French LM FlauBERT (Le et al., 2020). My contributions principally focus on biomedical WSD and biomedical Question Answering. I also evaluate the capacities of generative models pretrained on French data to perform WSD on general and biomedical data. I also contributed to the constitution of the pre-training corpus for the general domain version of FlauBERT2.
- **Coexya** Lyon, France | March 2022 - Sep. 2022
NLP Engineer (intern)
 - **End-of-studies Internship - ConSoRe:** I worked on a medical software called ConSoRe, dedicated to oncologists. I principally worked on the NLP Webservice. My goal was to extract and structure new medical concepts from medical reports. To do so I used the SpaCy library, SPARQL queries and the main programming languages I used were Python and Java.
- **SYNALP, LORIA** Nancy, France | May 2021 - Aug. 2021
Internship
 - **Research Internship:** I was in charge of developing a Web application, with Python Flask, where a teacher can create a learning path for its students by having some help to create some exercises. It can auto-generate questions targeting some grammatical notions that the teacher will previously choose. I also took care of the database redefinition and development, by using Flask SQLAlchemy. In a second part, I was in charge of designing a model specialized in *topic modeling* to determine main topics from the input texts. I had to create my own dataset from WikiData. I fine-tuned and used some BERT models (FlauBERT, CamemBERT, DistillBERT) pre-trained and designed by Hugging Face.

PROJECTS

- **OpenSemShift: Tracing and visualizing multilingual diachronic semantic change with contextualized embeddings:** Development of a web interface to visualize the semantic shift of a sample of words on different time periods. Development of different models for English and German.
- **Spoken Language Recognition:** Implementation and training, in PyTorch, of a CNN network to extract features from Mel Spectrograms images to decipher which is the spoken language.
- **NLP terms tagger:** Collaborative annotation of terms corresponding to the NLP domain in research articles. Development of an NLP terms tagger by finetuning different models from SpaCy.
- **End-of-studies Project - Polytech Grenoble and GETALP, LIG:** In collaboration with the GETALP team from the Grenoble Laboratory of Computer Science, my team and I worked on the "GenderedNews" project. We web-scraped newspaper data from Twitter. We redefined the pipeline to process the newspaper articles. We changed the data visualizations. We also worked on gender disambiguation mentioned in newspaper articles, using SpaCy.

PUBLICATIONS

- **Natural Language Inference for Clinical Trials (Aguiar, 2026)**: *PhD thesis*. Thesis to be defended in mid-November 2026.
- **Impact de l'affinage de modèles génératifs pour l'inférence en langue naturelle appliquée aux essais cliniques : comparaison avec des approches de few-shot learning (Kebdi et al., 2026)**: *Oral presentation at TALN 2026*. Comparison of finetuning and optimized few-shot approaches for NLI applied to clinical trials.
- **Assessing the Difficulty of Inference Types in Natural Language Inference for Clinical Trials (Aguiar et al., 2026)**: *Poster presentation at LREC 2026*. Fine-grained evaluation of the influence of inference types on LLMs' performance over the NLI4CT task.
- **Am I eligible? Natural Language Inference for Clinical Trial Patient Recruitment: the Patient's Point of View (Aguiar et al., 2025)**: *Oral presentation at CL4Health- NAACL 2025*. New NLI task proposing to tackle the patient-recruitment problem for clinical trials using patient language instead of using medical language to describe patients' profiles.
- **Constituting a dataset for applying Natural Language Inference to Chinese Clinical Trials: possible approaches and challenges (Aguiar et al., 2024)**: *Poster presented at Junior Conference on Data Sciences and Engineering 2024*. Presenting different approaches explored to build an NLI dataset for Chinese clinical trials. Work realized in collaboration with my M1 intern.
- **Récentes avancées de l'inférence en langue naturelle pour les essais cliniques (Aguiar et al., 2024)**: *Presented at Journée Santé & IA - PFIA 2024*. Literature review, following the PRISMA framework, tackling articles published in years 2022 and 2023 regarding NLI for clinical trials.
- **SEME at SemEval-2024 Task 2: Comparing Masked and Generative Language Models on Natural Language Inference for Clinical Trials (Aguiar et al., 2024)**: *Published at NAACL 2024 for the SemEval workshop*. 2 approaches proposed to solve the Natural Language Inference for Clinical Trials (NLI4CT) task. The first approach was based on finetuning and ensembling Masked Language Models, the second was based on prompting Flan-T5 using different prompting techniques.
- **Jargon: A Suite of Language Models and Evaluation Tasks for French Specialized Domains (Segonne et al., 2024)**: *Submitted during my M2 internship and published at LREC-COLING 2024*. Release and benchmark of domain-specific models (biomedical, legal, and written speech) for French.

SUPERVISION

- **Impact de l'affinage de modèles génératifs pour l'inférence en langue naturelle appliquée aux essais cliniques : comparaison avec des approches de few-shot learning** - Jan.-March, 2026: Supervision of 2 M1 students from the Master of Artificial Intelligence. We compared finetuning and few-shot approaches using LLMs for applying NLI to clinical trials.
- **Constituting a dataset for applying Natural Language Inference to Chinese Clinical Trials** - May-August, 2024: Supervision of an M1 student from the Master of Artificial Intelligence. We worked on constituting a novel dataset for applying NLI to Chinese clinical trials.

REVIEWING COMMITTEES

- **ACL Rolling Review** - May 2026
- **HeaLing - EACL 2026** - Dec. 2025
- **CL4Health - NAACL 2025** - Feb. 2025

VOLUNTEER EXPERIENCE

- **LISN's PhD students association co-founder** LISN, Orsay | Oct. 2024 - Now
Co-founded the PhD student association of my lab in order to organize scientific and non-scientific events between PhD students
- **LISN's PhD Days Organizer** LISN, Orsay | Feb. 2024 - April 2024
Organization of the CaLCUL conference (logistics, communication, caterer, etc.).
- **Chair of CaLCUL conference** IDMC, Nancy | Oct. 2022 - Feb. 2023
Organization of the CaLCUL conference (logistics, communication, managing EasyChair) and review of papers from NLP Master's M1 students
- **Tutor** Ark-en-Ciel, Vandoeuvre-lès-Nancy | Sep. 2018 - June 2019
I helped primary, middle, and high school students with homework and lessons at a non-profit association.
- **Partnerships Manager** BDA Polytech Nancy, Vandoeuvre-lès-Nancy | Sep. 2018 - June 2019
I was responsible for finding my school's Arts Bureau's sponsorships and partnerships to collaborate with various artistic and cultural organizations.